

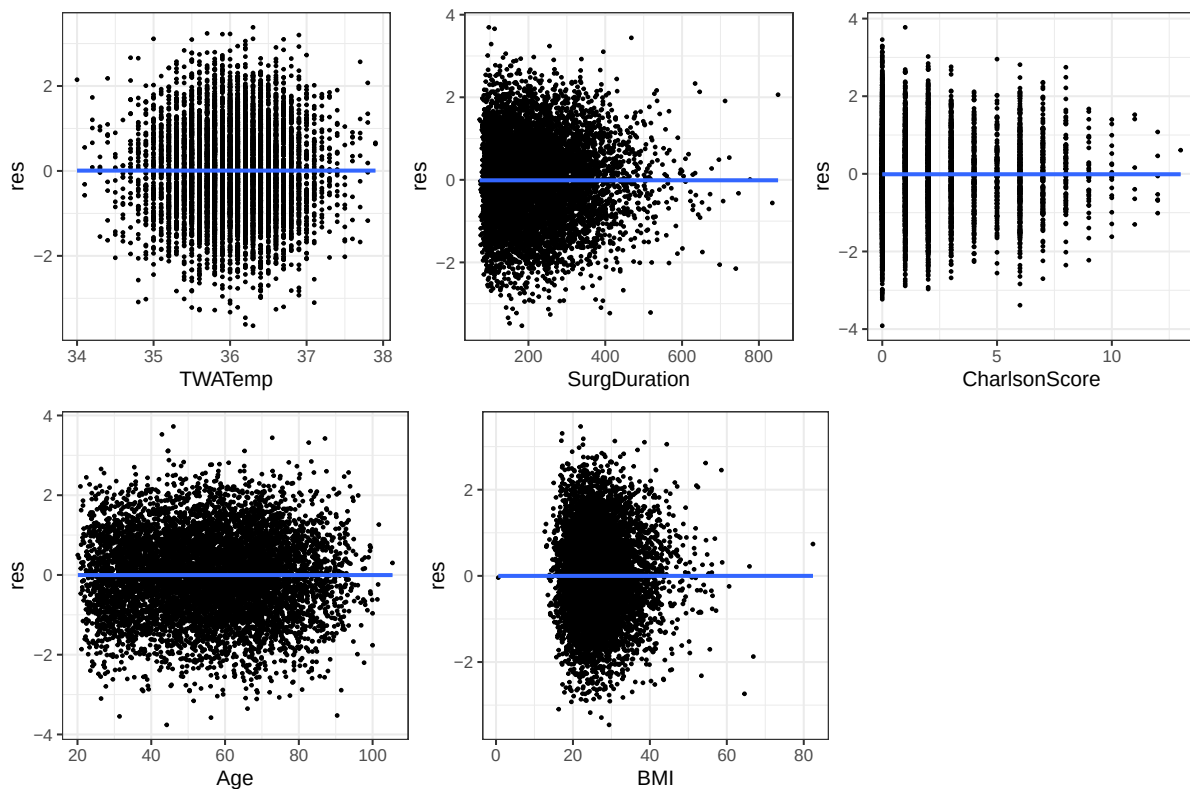
Model 2

Model 2

```
m2 <- glm(SeriousInfection ~ TWATemp + I(TWATemp^2) + SurgDuration + Open +  
  Age + FEMALE + BMI + CharlsonScore + CHF + VALVE +  
  RENLFAIL + LIVER + METS + TUMOR + COAG +  
  OBESE + LYLES + BLDLOSS + ANEMDEF + DRUG +  
  SteroidHx + ImmunosuppressantHx,  
  data = ct, family = binomial)
```

Diagnostics

Quantile residual plots:



Cook's distance:

```
summary(cooks.distance(m2))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.821e-07	2.601e-06	6.037e-06	1.295e-04	1.961e-05	7.202e-03

Variance inflation factors:

```
car::vif(m2)
```

TWATemp	I(TWATemp^2)	SurgDuration	Open
7487.988855	7488.054313	1.074024	1.043158
Age	FEMALE	BMI	CharlsonScore
1.396642	1.035091	1.676480	7.193757
CHF	VALVE	RENLFAIL	LIVER
1.453500	1.286886	1.783227	1.152349
METS	TUMOR	COAG	OBESE
4.103702	2.055072	1.173089	1.617495
LYTES	BLDLOSS	ANEMDEF	DRUG
1.167232	1.026627	1.112369	1.043884
SteroidHx	ImmunosuppressantHx		
1.092130	1.041770		

Model summary

```
summary(m2)
```

Call:

```
glm(formula = SeriousInfection ~ TWATemp + I(TWATemp^2) + SurgDuration +  
  Open + Age + FEMALE + BMI + CharlsonScore + CHF + VALVE +  
  RENLFAIL + LIVER + METS + TUMOR + COAG + OBES + LYTES +  
  BLDLOSS + ANEMDEF + DRUG + SteroidHx + ImmunosuppressantHx,  
  family = binomial, data = ct)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.029e+02	1.127e+02	2.688	0.00720 **

TWATemp	-1.711e+01	6.253e+00	-2.735	0.00623	**
I(TWATemp^2)	2.389e-01	8.675e-02	2.754	0.00588	**
SurgDuration	2.975e-03	3.647e-04	8.156	3.46e-16	***
Open	1.929e-01	1.185e-01	1.628	0.10361	
Age	-6.831e-03	2.876e-03	-2.376	0.01752	*
FEMALE	-1.025e-01	8.348e-02	-1.228	0.21933	
BMI	2.385e-03	8.515e-03	0.280	0.77944	
CharlsonScore	7.024e-02	5.006e-02	1.403	0.16057	
CHF	2.037e-01	2.225e-01	0.915	0.36006	
VALVE	-1.820e-01	1.871e-01	-0.973	0.33068	
RENLFAIL	2.221e-01	2.164e-01	1.026	0.30471	
LIVER	-4.626e-01	2.741e-01	-1.688	0.09147	.
METS	-6.303e-01	3.441e-01	-1.832	0.06700	.
TUMOR	-2.119e-01	1.534e-01	-1.381	0.16729	
COAG	4.692e-01	1.318e-01	3.561	0.00037	***
OBESE	-2.270e-01	1.410e-01	-1.610	0.10740	
LYTES	6.025e-01	8.914e-02	6.759	1.39e-11	***
BLDLOSS	1.804e-01	2.354e-01	0.766	0.44359	
ANEMDEF	3.035e-02	9.682e-02	0.313	0.75392	
DRUG	5.422e-01	2.240e-01	2.420	0.01551	*
SteroidHx	6.106e-02	8.897e-02	0.686	0.49256	
ImmunosuppressantHx	-9.748e-02	2.315e-01	-0.421	0.67372	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4608.2 on 7907 degrees of freedom
 Residual deviance: 4386.0 on 7885 degrees of freedom
 AIC: 4432

Number of Fisher Scoring iterations: 5